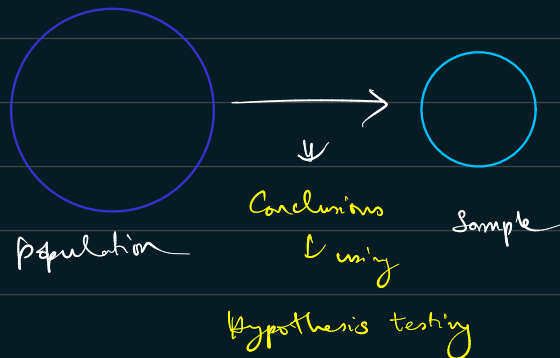


* Hypothesis testing :-

→ Part of inferential stats.

→ to arrive at conclusions / inference



* Hypothesis testing Mechanism:-

1. Null Hypothesis (H_0) :-

The assumption you are beginning with

2. Alternate Hypothesis (H_1)

The opposite of null hypothesis.

3. Experiments → Statistical Analysis

4. Accept / Reject the null hypothesis

Ex:- Person in a court judgment.

H_0 = The person is not guilty until proven

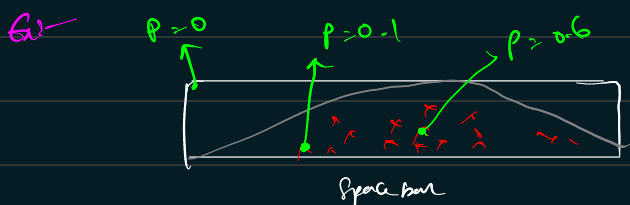
H_1 = The person is guilty

Collect proofs (finger prints, etc.)

* P value :-

→ P value is a number, calculated from a statistical test, that describes how likely you are to have found a particular set of observations if the null hypothesis were true.

→ P values are used in hypothesis testing to help decide whether to reject the null hypothesis.



→ Out of 100 touches, we touch around 10 times at test region ($p=0.1$)

Ex:- Coin is fair or not

for 100 times

$$P(H) = 0.5$$

$$P(T) = 0.5$$

→ fair

$$P(H) = 0.6$$

$$P(T) = 0.4$$

↓

$$P(H) = 0.7$$

$$P(T) = 0.3$$

→



need to perform hypothesis testing.

① Null hypothesis -

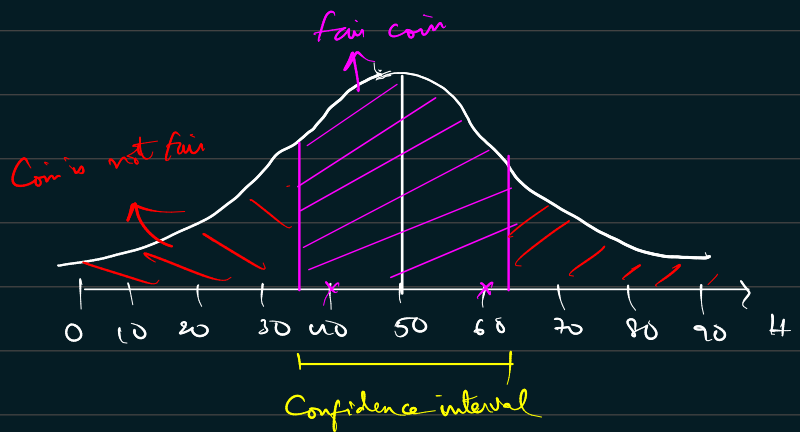
H_0 = Coin is fair.

② Alternate hypothesis.

H_1 = Coin is not fair

③ Experiment

Toss it for 100 times



To find C.I.



Significance interval (α)

$$\alpha = 0.05$$

$$CI = 1 - 0.05 = 0.95$$

→ If $p < \text{Significance value}$ → reject null hypothesis

* Hypothesis testing and Statistical Analysis:-

1. Z test } mean $\rightarrow$ Z table $\rightarrow$ Z score, p value
2. t test } $\rightarrow$ t table
3. Chi Square test $\rightarrow$ Categorical data
4. Anova test $\rightarrow$ Variance

1. Z test:- a) pop std b) $n > 30$

Ex

The average heights of all residents in a city is 168cm with sd 3.9. A doctor believes the mean to be different. He measured the height of 36 individuals and found the average height to be 169.5cm.

- a) State null and alternate hypothesis
- b) At a 95% confidence level, is there enough evidence to reject the null hypothesis?

Sol

$$\mu = 168 \text{ cm}$$

$$n = 36 > 30$$

Can use Z-test

$$\sigma = 3.9$$

$$\bar{x} = 169.5 \text{ cm}$$

$$CF = 95\%$$

$$\text{Significance interval } (\alpha) = 1 - CF$$

$$= 1 - 0.95 = 0.05$$

① Null Hypothesis:-

$$H_0 = \mu = 168 \text{ cm}$$

② Alternate Hypothesis

$$H_1 = \mu \neq 168 \text{ cm}$$

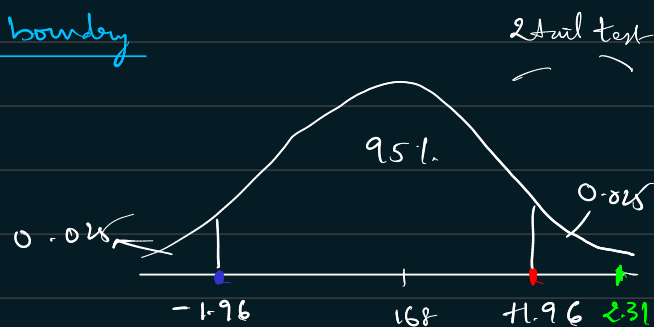
③ Based on CF we draw Decision boundary

Total area = 1



$$1 - 0.025 = 0.975 \Rightarrow Z \text{ score}$$

$$= 1.96$$



→ If Z_{computed} is less than -1.96 or greater than $+1.96$, we reject the null hypothesis.

Z-test:

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$Z = \frac{\bar{x} - \mu}{\sigma} \rightarrow \text{for population}$$

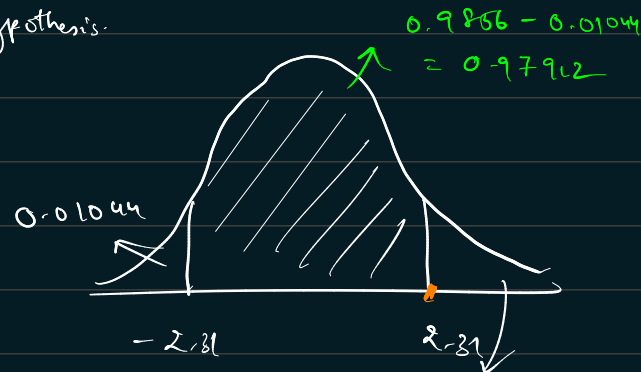
$\sigma/\sqrt{n}$ for sample

$$Z = \frac{169.5 - 168}{39/\sqrt{36}}$$

$$= \frac{1.5}{0.65} = 2.31$$

$2.31 > 1.96 \Rightarrow$ Reject null hypothesis.

0.9856



$$p \text{ value} = 0.01044 + 0.01044$$

$$= 0.02088$$

$$1 - 0.9856$$

$$= 0.01044$$

$$p < 0.05$$

$0.02088 < 0.05 \Rightarrow$ Reject null hypothesis

The avg height $\neq 168$ cm but > 168 cm.

- ⑤ A factory manufactures bulbs with an average warranty of 5 years with sd of 0.50. A worker believes that the bulb will malfunction in less than 5 years. He tests a sample of 40 bulbs and finds the average time to be 4.8 years.

a) State null and alternate hypothesis

b) At a 2% significance level, Is there enough evidence to support his idea that the warranty should be revised?

SL

$$\mu = 5 \text{ years}$$

$$\sigma = 0.50$$

$$\text{Significance level } (\alpha) = 0.02$$

$$n = 40 > 30$$

$$\bar{x} = 4.8$$

$$\alpha = 1 - c\alpha$$

$$c\alpha = 1 - 0.02$$

$$= 0.98$$

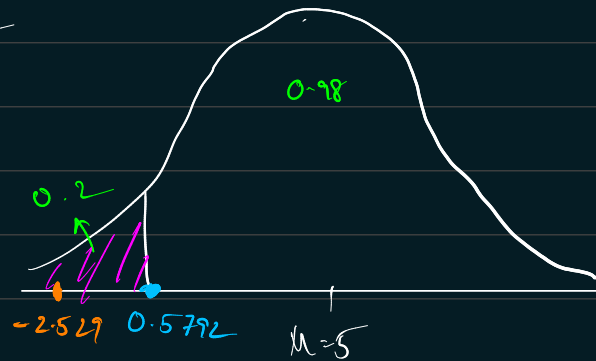
① Null Hypothesis:-

$$H_0 = \mu = 5$$

② Alternate Hypothesis:-

$$H_1 = \mu < 5 \rightarrow \text{1 tail test}$$

③ Decision boundary:-



$$Z_{\text{critical}} \rightarrow 1 - 0.98 = 0.02$$

$$Z_{\text{critical}} = 0.5792$$

$$Z_{\text{cal}} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{4.8 - 5}{0.5/\sqrt{40}} = \frac{-0.2}{0.079} = -2.529$$

$\therefore$ Reject H_0 & At 99% A.I. $c\alpha$ $\mu < 5$ yrs

3) A factory has a machine that dispenses 80 ml of fluid in a bottle. An employee believes the average amount of fluid is not 80ml. Using 40 samples, he measures the average amount dispensed by the machine to be 78ml with a standard deviation of 2.5.

a) State the null and alternate hypothesis.

b) At a 95% confidence level, is there enough evidence to support the idea that the machine is not working properly?

Sol

$$\mu = 80 \text{ ml}$$

$$\sigma = 2.5$$

$$n = 40 > 30$$

$$\bar{x} = 78 \text{ ml}$$

$$H_0 = \mu = 80$$

$$H_1 = \mu \neq 80 \rightarrow \text{2-tail test.}$$

$$1 - 0.025 = 0.975$$

$$Z_{\text{critical}} = 1.96$$



$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$= \frac{78 - 80}{2.5 / \sqrt{40}} = \frac{-2}{0.3952} = -5.059 < -1.96$$

Reject $H_0 \rightarrow \mu \neq 80$ at 95%.

* Student t distribution :-

→ When σ is not available

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

1.

In a population the average IQ is 100. A team of researchers want to test a new medication to see if it has either a positive or negative effect on intelligence, or no effect at all. A sample of 30 participants who have taken the medication has a mean of 140 with a sd of 20. Did the medication affect intelligence?

2d

$$\mu = 100$$

$$n = 30$$

$$[CI = 95\%]$$

$$\bar{x} = 140$$

$$s = 20$$

$$\alpha = 0.05$$

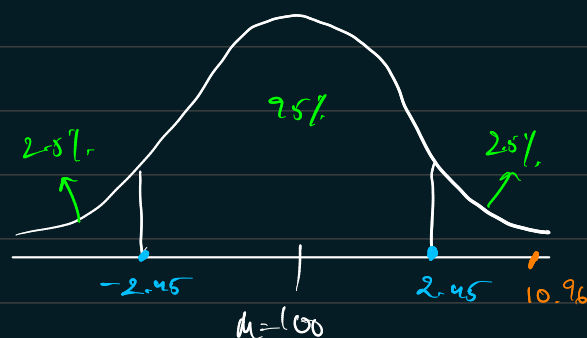
$$H_0: \mu = 100$$

$$H_1: \mu \neq 100 \rightarrow 2\text{tail test}$$

$$\text{Degree of freedom, } df = n - 1 \\ = 30 - 1 = 29$$

[From t table :-
Against α value & dof]

$$t_{\text{critical}} = 2.45$$



Calculate test statistic :-

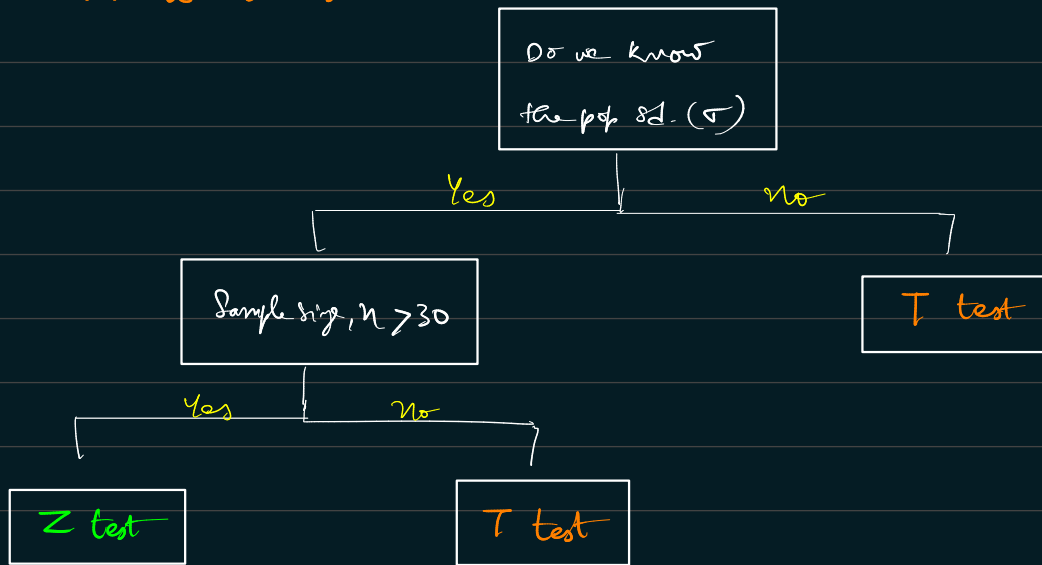
$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$t_{\text{cal}} = \frac{140 - 100}{20/\sqrt{30}} = 10.96 > 2.45$$

Reject H_0

Medication has effect on intelligence (in case, > 100)

* Z test vs t-test :-



* Type 1 & Type 2 Errors :-

Reality :- Null Hypothesis is True or False

Decision :- Null Hypothesis is True or False

Outcome 1 :- We reject the null hypothesis when in reality it is false
→ Good

Outcome 2 :- We reject the null hypothesis when in reality it is True
→ Type 1 error

Outcome 3 :- We accept the null hypothesis when in reality it is False.
→ Type 2 error

Outcome 4 :- We accept the null hypothesis when in reality it is True.
→ Good

		Decision	
		True	False
Reality	True	Good	Type 1
	False	Type 2	Good

* Bayes Statistics

Bayesian statistics is an approach to data analysis and parameter estimation based on Baye's theorem.

Independent Events

Ex: Rolling a dice.

$\{1, 2, 3, 4, 5, 6\}$

$$P(1) = \frac{1}{6}$$

$$P(2) = \frac{1}{6}, \dots$$

→ One event doesn't impact another

Dependent Events

Ex:-



$$P(R) = \frac{2}{5} \rightarrow P(Y) = \frac{3}{5}$$

$$P(R \text{ and } Y) = P(R) \cdot P(Y/R)$$

↓
Conditional Probability

$$= \frac{2}{5} \cdot \frac{3}{4} = \frac{6}{20}$$

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) \cdot P(B/A) = P(B) \cdot P(A/B)$$

$$P(A/B) = \frac{P(A) \cdot P(B/A)}{P(B)}$$

→ Baye's theorem

$A, B \rightarrow$ events

$P(A), P(B) \rightarrow$ Independent prob's of A & B

$P(A/B) \rightarrow$ Prob. of A given B is true

Ex: House price prediction

Size of house

No. of rooms

Location

Price

x_1

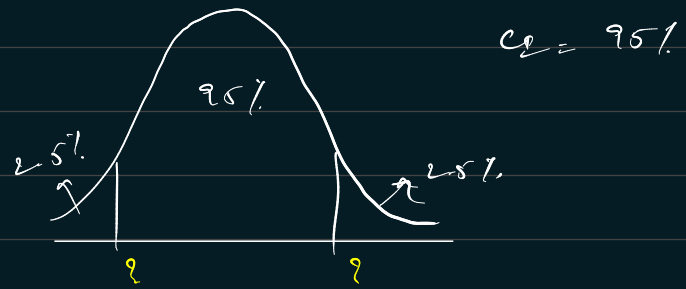
x_2

x_3

y

$$P(y/x_1, x_2, x_3) = \frac{P(y) \cdot P(x_1, x_2, x_3/y)}{P(x_1, x_2, x_3)}$$

* Confidence interval and margin of error:-



Point estimate :-

$\bar{x} \longrightarrow \bar{\mu}$ (To estimate pop. mean from sample mean)

2.5 Actual $\mu = 3$

2.7 $\longrightarrow$? (doesn't provide any range)

3.2

3.6

Confidence interval :-

Point estimate $\pm$ margin of error.

for Z test :-

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$t_{\alpha/2}$ for test.

Ex:-

In an exam, the sd of score is 100. A sample of 30 test takers have a mean score of 520. Construct 95% CI about mean.

Sol

$$\sigma = 100$$

$$n = 30$$

$$\bar{x} = 520$$

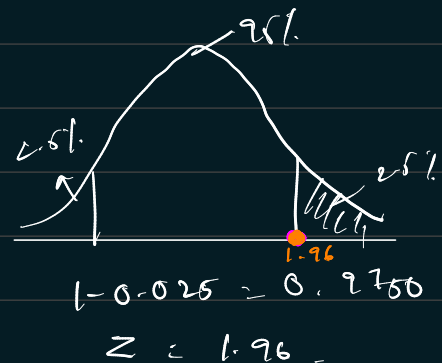
$$CI = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\alpha = 1 - CI$$

$$= 1 - 0.95$$

$$= 0.05$$

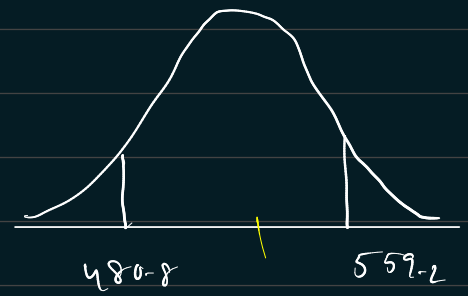
$$CI = 520 \pm$$



$$\text{lower CI} = 520 - (1.96) \frac{100}{\sqrt{25}} \approx 480.8$$

$$\text{higher CI} \approx 520 + (1.96) \frac{100}{\sqrt{25}} \approx 559.2$$

⇒ 95% Confident about the mean score is
b/w 480.8 & 559.2



Q

The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a sd of 3.

a) State null and alternate hypothesis

b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? *[use p-value method]*

Sol

$$\mu = 168 \text{ lbs}$$

$$n = 36 > 30$$

$$CI = 95\%$$

$$\bar{x} = 169.5$$

$$\alpha = 1 - CI$$

$$s = 3.9$$

$$= 1 - 0.95$$

$$= 0.05$$

$$H_0 = 168 \text{ lbs}$$

$$H_1 \neq 168 \text{ lbs} \rightarrow 2 \text{ tail test}$$

Z test

$$1 - 0.025$$

$$= 0.975$$

$$Z_{0.975} = 1.96$$



$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}} = \frac{1.5}{0.65}$$

$$Z = 2.31 > 1.96$$

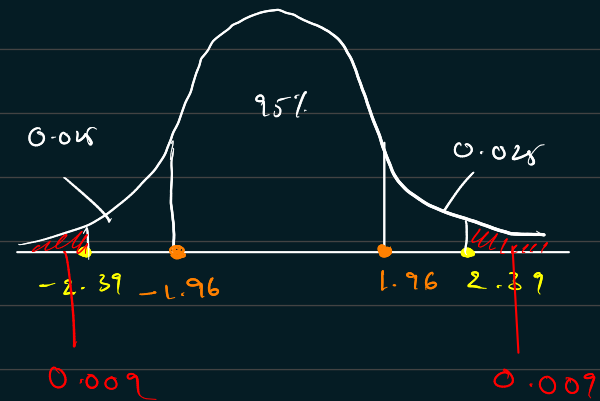
$\Rightarrow$ Reject H_0

p-value method :-

p value $\rightarrow$ area of the red shaded region

$$Z_{0.99} = 2.39$$

$$\text{Area of red shaded region} = 1 - 0.991 \\ = 0.009$$



$$1 - 2(0.009) = 0.982 \\ = 98.2\%$$

$$p\text{-value} = 2(0.009) \\ = 0.018 < \alpha = 0.05$$

p-value $< \alpha \rightarrow$ Reject H_0

p-value $\geq \alpha \rightarrow$ Accept H_0

Q. A factory manufactures cars with a warranty of 5 years on the engine and transmission. An engineer believes that the engine or transmission will malfunction in less than 5 years. He tests a sample of 40 cars and find the average time to be 4.8 years with a sd of 0.5.

a) State the null and alternate hypothesis

b) At a 2% significance level, is there enough evidence to support the idea that the warranty should be revised?

Sol

$$\mu = 5 \text{ years}$$

$$n = 40$$

$$\alpha = 0.02$$

$$\bar{x} = 4.8 \text{ yrs}$$

$$CR = 1 - \alpha$$

$$s = 0.5$$

$$= 98\%$$

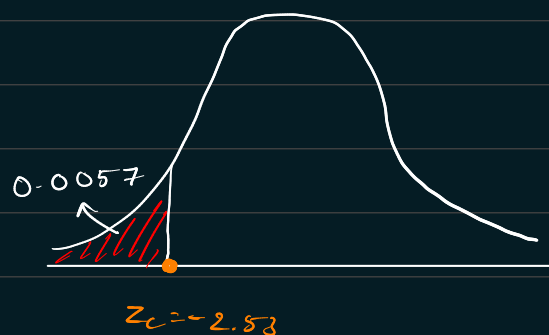
$$H_0: \mu \geq 5$$

$$H_1: \mu < 5 \rightarrow \text{1-tail test}$$

$$Z_{\text{critical}} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$Z_c = \frac{4.8 - 5}{0.5/\sqrt{40}} = \frac{-0.2}{0.079}$$

$$Z_c = -2.53$$



from $Z_c \rightarrow$ Area of red shaded region

$$p \text{ value} = 0.0057 < \alpha = 0.02$$

$$p < \alpha$$

$\Rightarrow$ Reject H_0

